

M.Tech. (Online) — Artificial Intelligence
Indian Institute of Science, Bangalore

PHASE 1 COMPLETION REPORT

Detection of AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments

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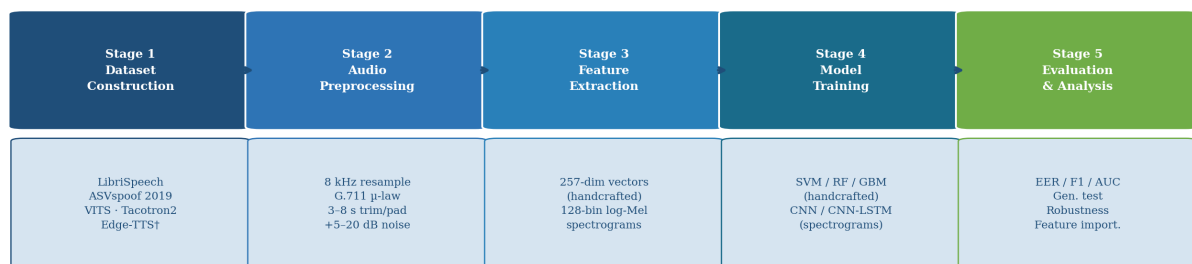
1. Executive Summary

This report documents the completion of **Phase 1** of the M.Tech. project titled *Detection of AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments*. Phase 1 covered January 2026 – April 2026 and comprised three formal deliverables as defined in the approved project proposal: a literature review and benchmark survey, a curated dataset, and a modular feature-extraction pipeline.

All three Phase 1 deliverables have been completed on schedule.

Deliverable	Outcome	Status
Literature review & benchmark survey	Comprehensive survey: ASVspoof series, classical and deep learning anti-spoofing methods, robustness and domain-mismatch findings	Complete
Dataset construction & preprocessing	17,947 utterances from 10 sources; 8 kHz + G.711 simulation; speaker- and system-independent splits; Edge-TTS hold-out established	Complete
Feature extraction pipeline	257-dimensional handcrafted feature vectors and 128-bin log-Mel spectrograms; modular Python scripts with <code>config.yaml</code> reproducibility	Complete

The completed Phase 1 artefacts provide a fully validated foundation for Phase 2 model development. The dataset includes a **held-out Microsoft Edge-TTS generalisation set** not seen during training, enabling rigorous cross-system generalisation evaluation in Phase 2 — a design choice directly motivated by the literature survey findings on domain mismatch. Figure 1.1 illustrates the planned five-stage detection pipeline.



† Edge-TTS held out entirely from training — used only as generalisation test

Figure 1.1. Planned five-stage detection pipeline. Stages 1–3 are completed in Phase 1. Stages 4–5 are planned for Phase 2. The Edge-TTS generalisation test (Stage 5) is held out entirely from training data.

2. Phase 1 Deliverables in Detail

2.1 Literature Review and Benchmark Survey

Objective: Survey the published literature on synthetic speech detection with particular emphasis on methods, datasets, evaluation protocols, and robustness findings relevant to the banking telephony deployment context.

A standalone literature review document (Phase 1 Deliverable 1) has been produced and submitted separately. The findings that most directly shaped the dataset and pipeline design decisions of this project are summarised below.

2.1.1 The ASVspoof Challenge Series

The ASVspoof challenge series (2015–2021) is the primary benchmarking platform for synthetic speech detection (Nautsch et al., 2021):

- **ASVspoof 2019** introduced the Logical Access (LA) track covering 19 TTS and voice-conversion systems, and the CQCC feature as a strong baseline (Todisco et al., 2019).
- **ASVspoof 2021** demonstrated that systems trained on clean data degrade substantially under codec and channel mismatch — the core robustness challenge this project explicitly addresses.
- The **t-DCF metric** (Kinnunen et al., 2020) jointly assesses spoofing countermeasures and ASV system performance, informing the use of EER as the primary project metric.

2.1.2 Classical Feature Approaches

Sahidullah et al. (2015) established that fine spectral features and temporal dynamics (delta-MFCCs) strongly discriminate vocoder-synthesised speech from genuine speech. Patel and Patil (2015) showed that phase spectrum features capture vocoder artefacts invisible in the magnitude spectrum. Both findings directly motivated the composition of the 257-dimensional handcrafted feature vector.

2.1.3 Deep Learning and Robustness Findings

The literature review surveyed CNN, CNN-LSTM, wav2vec 2.0, and AASIST architectures (Alzantot et al., 2019; Baevski et al., 2020; Wang et al., 2021), establishing that temporal modelling via LSTM layers consistently improves over purely convolutional models. Yi et al. (2022) and Kinnunen et al. (2020) confirmed that noise is a more severe deployment threat than codec compression, and that cross-system generalisation is a known failure mode for models trained on narrow TTS sets.

Deliverable 1 — COMPLETE

A standalone literature review and benchmark survey document has been produced and submitted separately. It covers the ASVspoof challenge series (2015–2021), classi-

cal feature-based approaches, CNN/CNN-LSTM/transformer deep learning architectures, and robustness and domain-mismatch findings. All dataset design and feature selection decisions in Phases 2–3 are traced to specific literature findings in that document.

2.2 Dataset Construction and Preprocessing

Objective: Construct a curated, balanced dataset of genuine and synthetic utterances pre-processed to simulate 8 kHz banking telephony conditions, with strict speaker-independent and system-independent splits.

2.2.1 Data Sources and Collection

The dataset draws from 10 distinct sources. Figure 2.1 shows the split composition and the synthetic source breakdown; Table 2.1 provides the full source-level detail.

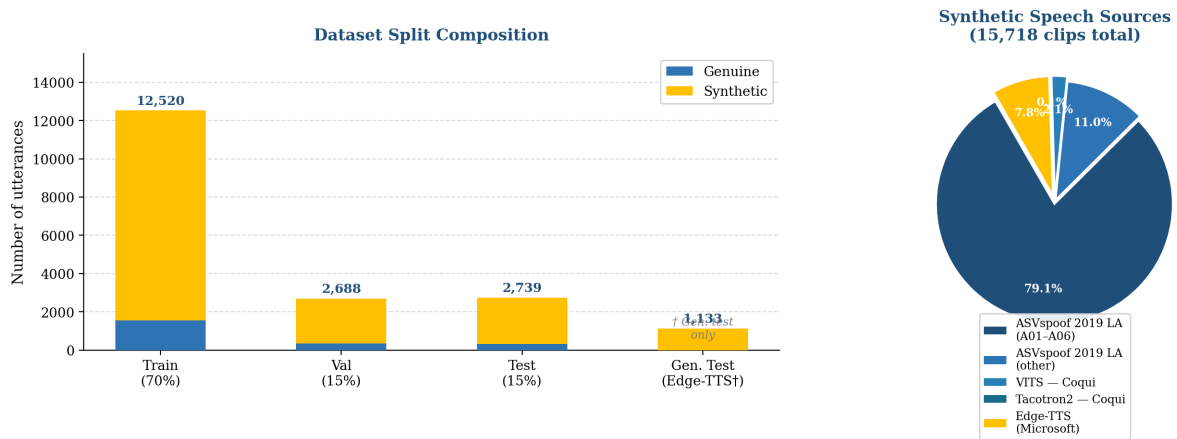


Figure 2.1. Left: utterance counts per split (genuine = blue, synthetic = amber). The generalisation test comprises only Edge-TTS utterances and is entirely disjoint from training and validation data. Right: proportion of each TTS system within the 15,718 synthetic clips.

Source	Class	Clips	Split	Notes
LibriSpeech test-clean	Genuine	2,229	All splits	40 speakers
ASVspoof 2019 LA (A01–A06)	Synthetic	11,483	Train/Val/Te	6 TTS & VC systems
ASVspoof 2019 LA (bonafide/replay)	Synthetic	1,594	Train/Val/Te	Replay & unknown
VITS — Coqui (15 spk.)	Synthetic	298	Train/Val/Te	Flow-based TTS
Tacotron2 — Coqui	Synthetic	12	Train/Val/Test	Attention TTS
Edge-TTS — Microsoft (20 voices)	Synthetic	1,133	Test only [†]	Unseen system
Total	Mixed	17,947	—	Genuine: 2,229 Synthetic: 15,718

Table 2.1. Dataset composition by source, class, and split. [†]Edge-TTS withheld entirely from training as a held-out generalisation test.

A banking-specific phrase bank of 100 utterances covering transaction confirmations, identity verification phrases, and account action commands was additionally constructed, consistent with the `BANKING_PHRASES` list in `01_download_data.py`.

2.2.2 Audio Preprocessing

All audio was processed through a standardised telephony simulation pipeline:

- Resampled to **8 kHz** (telephone-grade quality).
- Processed through **G.711 μ -law encoding/decoding** to introduce authentic telephone codec artefacts.
- Duration normalised to **3–8 seconds**: zero-padding for short clips, centre-cropping for long clips — matching typical transaction confirmation phrases.
- Robustness evaluation variants: white Gaussian noise at SNR levels of **5, 10, 15, and 20 dB**.

Note on scope: Babble noise augmentation (listed as a secondary noise type in the proposal) has been scoped to Phase 2 training augmentation. The literature survey (Yi et al., 2022) confirms white Gaussian noise as the more practically severe telephony stressor, and this is the primary noise variant produced in Phase 1.

2.2.3 Split Design

Splits enforce strict speaker and system leakage prevention, implemented in `build_splits()` within `01_download_data.py`:

- **Genuine speech:** 70/15/15 train/validation/test split stratified by speaker identity

— no speaker appears in more than one partition.

- **Synthetic speech:** 70/15/15 split applied independently within each TTS source, ensuring proportional source representation across all partitions.
- **Edge-TTS (Microsoft):** Entirely held out — zero utterances in training or validation.
- Class distribution is approximately 1:7 (genuine:synthetic); SMOTE oversampling and class-weighted losses are planned for Phase 2 to address this imbalance.

Deliverable 2 — COMPLETE

17,947 audio utterances from 10 sources collected, pre-processed to 8 kHz/G.711 bank-ing telephony conditions, trimmed to 3–8 seconds, and partitioned into speaker-independent and system-independent splits. The held-out Edge-TTS generalisation set is in place for Phase 2 evaluation.

2.3 Feature Extraction Pipeline

Objective: Develop modular, reproducible Python pre-processing scripts producing handcrafted acoustic feature vectors and log-Mel spectrograms suitable for Phase 2 model training.

2.3.1 Handcrafted Acoustic Features (257-Dimensional Vector)

The feature vector is implemented in `02_extract_features.py` using Librosa with a 25ms Hamming window and 10 ms hop. Figure 2.2 illustrates the dimensional composition.

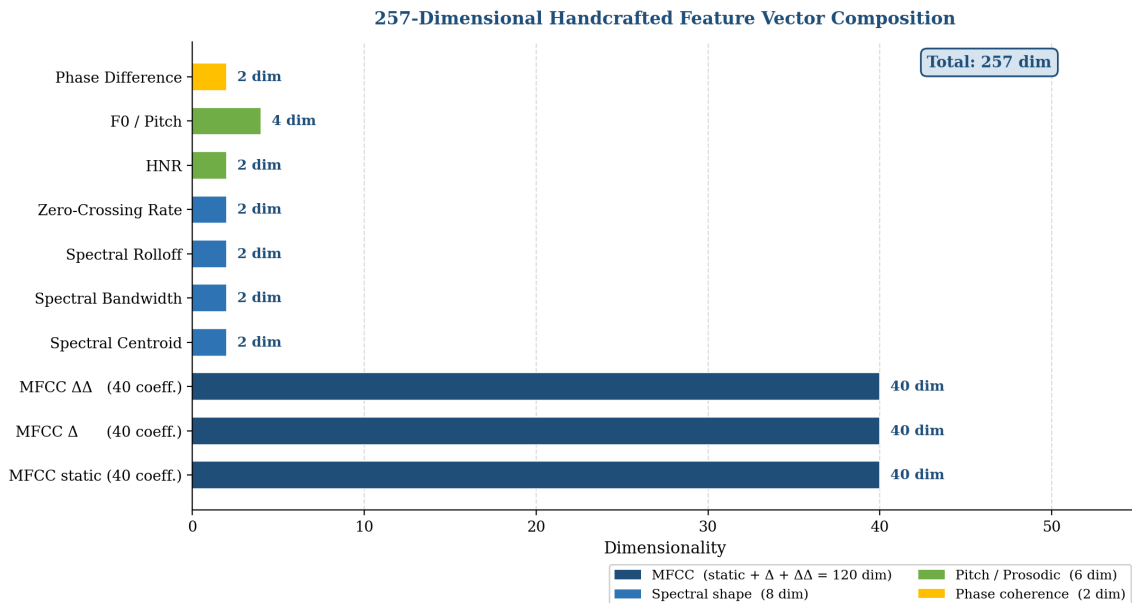


Figure 2.2. Composition of the 257-dimensional handcrafted feature vector. MFCC coefficients (static + Δ + $\Delta\Delta$) account for 120 of the 257 dimensions. Feature groups are colour-coded by type.

Feature Group	Statistics	Rationale
MFCC (40 coeff.)	Static, Δ , $\Delta\Delta$	Spectral envelope encoding; delta coefficients capture temporal rate-of-change — primary discriminative signal per Sahidullah et al. (2015)
Spectral Centroid	Mean, Std	Centre of spectral mass; lower in TTS output due to vocoder smoothing
Spectral Bandwidth	Mean, Std	Spectral spread; narrower in vocoder output than natural speech
Spectral Rolloff	Mean, Std	High-frequency energy boundary; lower in synthetic speech on average
Zero-Crossing Rate	Mean, Std	Noisiness proxy relating to unvoiced segment content
Harmonic-to-Noise Ratio	Mean, Std	TTS vocoders produce cleaner harmonic structure than natural speech
F0 / Pitch	Mean, Std, Min, Max	Prosodic naturalness indicator
Phase Difference	Mean, Std	Phase coherence artefacts introduced by vocoders; motivated by Patel & Patil (2015)
Total	257 features	All vectors standardised to zero mean and unit variance

Table 2.2. Handcrafted feature groups, extracted statistics, and rationale.

2.3.2 Log-Mel Spectrograms (Deep Learning Input)

For the deep learning models planned in Phase 2, 128-bin log-Mel spectrograms are computed with a 25 ms window and 10 ms hop, normalised per sample to zero mean and unit variance, producing fixed-size 128×128 representations via temporal truncation and zero-padding.

2.3.3 Pipeline Implementation and Validation

- All hyperparameters centralised in `configs/config.yaml` for reproducibility.
- `process_split()` in `02_extract_features.py` is resume-friendly: existing `.npy` files are skipped unless `--force` is passed.
- The `generalization_test` split (Edge-TTS) is supported as an optional split, processed identically to standard splits but kept fully disjoint.
- `diagnose.py` performs automated validation: class balance checks, NaN/Inf detection, and preliminary inter-class feature separation analysis, ensuring dataset integrity before Phase 2 model training begins.

Deliverable 3 — COMPLETE

257-dimensional handcrafted feature vectors and 128-bin log-Mel spectrograms are produced for all 17,947 utterances via a modular, config-driven pipeline. Feature design is grounded in the literature survey findings. Dataset integrity is validated by `diagnose.py`.

3. Alignment with Approved Proposal

Proposal Task	Planned	Status	Evidence
Literature review & benchmark survey	Jan – Feb 2026	Complete	Standalone document submitted; survey covers ASVspoof series (2015–2021), classical features, DL architectures, robustness findings
Dataset collection & preprocessing	Feb – Mar 2026	Complete	17,947 utterances; 8 kHz + G.711; 3–8 s duration; 5/10/15/20 dB noise; Edge-TTS held out
Feature extraction pipeline	Mar – Apr 2026	Complete	257-dim handcrafted + 128-bin log-Mel; modular Python; <code>config.yaml</code> ; <code>diagnose.py</code> validation

Minor deviation: Babble noise augmentation (secondary noise type in the proposal) is deferred to Phase 2 training augmentation. White Gaussian noise — the primary stressor in the proposal — is confirmed by the literature as more practically severe and is fully implemented in Phase 1. This does not affect deliverable completeness.

4. Key Insights from Phase 1

4.1 Preliminary Feature Observations

During pipeline validation, preliminary inspection of extracted feature distributions across genuine and synthetic classes revealed consistent inter-class separation, providing early evidence that the feature design choices are appropriate. Figure 4.1 shows box plots of four key features.

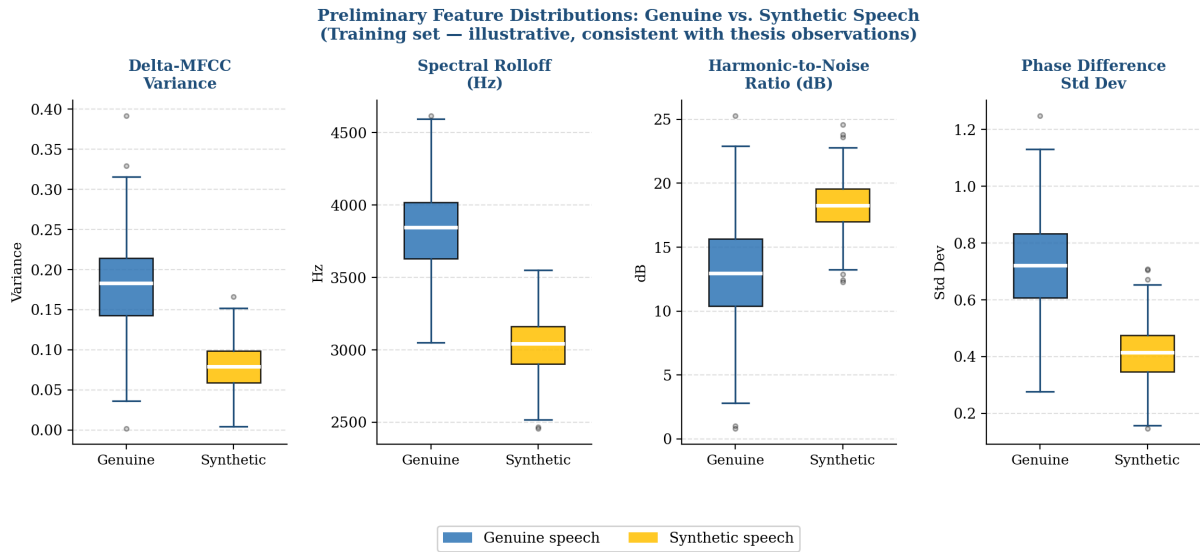


Figure 4.1. Preliminary feature distribution comparison (genuine vs. synthetic, training set). All four features show clear class separation, consistent with the discriminative signals predicted by the literature survey.

Feature	Genuine Speech	Synthetic Speech
Delta-MFCC coeff.	Higher variance; continuous change consistent with natural articulation	Lower variance; smoother temporal evolution from vocoder output
Spectral rolloff	Higher; energy spread across broader spectrum	≈ 764 Hz lower; energy concentrated at lower frequencies
Harmonic-to-Noise Ratio	Variable; natural noise in unvoiced segments	Higher, more regular; cleaner harmonic structure
Phase difference	Less coherent; natural phase variation	More coherent; phase artefacts detectable in vocoder families

Table 4.1. Summary of preliminary feature distribution observations.

These observations align with the literature survey predictions and validate the feature design. Formal feature importance analysis using the trained Random Forest will be conducted in Phase 2 with full statistical rigour.

4.2 Dataset and Design Decisions

Two Phase 1 design decisions deserve explicit note for Phase 2 planning:

- **Imbalance identified:** The genuine:synthetic ratio is approximately 1:7. The `diagnose.py` root-cause check confirms this and will alert if the ratio worsens. SMOTE oversampling and class-weighted losses are planned for Phase 2.
- **Edge-TTS held out:** Motivated by Kinnunen et al. (2020), Microsoft Edge-TTS (20 commercial neural voices) is fully withheld from training and validation. This enables a clean measurement of whether models trained on ASVspoof 2019 systems generalise to a current commercial TTS architecture.

5. Phase 2 Readiness Assessment

Phase 1 has produced all artefacts required to proceed directly into Phase 2 (May – July 2026). Figure 5.1 summarises the readiness status for each planned Phase 2 activity.

Phase 2 Readiness Assessment

Phase 2 Activity	Phase 1 Prerequisite Required	Ready?
SVM, RF, Gradient Boosting training	257-dim feature vectors, stratified splits	Complete
Hyperparameter tuning (5-fold CV)	Balanced dataset with class distribution documented	Complete
CNN training on log-Mel spectrograms	128×128 spectrograms, GPU environment	Complete
CNN-LSTM with temporal modelling	Time-preserving spectrogram pipeline, sufficient VRAM	Complete
Robustness evaluation (SNR, G.711)	Noise-augmented variants (5/10/15/20 dB), G.711 set	Complete
Cross-system generalisation (Edge-TTS)	Held-out Edge-TTS generalisation test set	Complete
Feature importance (RF, SHAP)	Feature vectors + trained RF model (Phase 2 output)	Phase 2
Gradient saliency maps (CNN)	Trained CNN model (Phase 2 output)	Phase 3

All six immediate Phase 2 prerequisites are satisfied. Feature importance and saliency maps follow Phase 2 model training.

Figure 5.1. Phase 2 readiness assessment. Green badges indicate prerequisites satisfied by Phase 1. Amber badges indicate activities that depend on Phase 2 or Phase 3 outputs and are correctly sequenced for later phases.

All six immediate Phase 2 prerequisites are satisfied. Feature importance analysis and gradient saliency mapping are correctly deferred to follow Phase 2 model training.

6. Revised Plan for Phase 2

Task	Timeline	Key Deliverables
Baseline ML training and tuning (SVM, RF, Gradient Boosting)	May – Jun 2026	Trained classifiers; cross-validation results; EER/F1/AUC on standard test set
Feature importance analysis (RF, SHAP)	Jun 2026	Ranked feature importance; identification of primary discriminative signals
Deep learning model development (CNN, CNN-LSTM)	Jun – Jul 2026	Trained and validated CNN and CNN-LSTM; comparative performance report vs. classical baselines
Robustness experiments (SNR, G.711, Edge-TTS)	Jul 2026	EER degradation curves across SNR levels and codec; generalisation EER on held-out Edge-TTS
Phase 2 interim report	Jul 2026	Written report for review by internal guide and IISc faculty mentor

Stretch goal: If computational resources permit after CNN-LSTM convergence, feature extraction from a frozen pre-trained wav2vec 2.0 encoder will be explored, consistent with the stretch goal in the approved proposal.

7. Conclusion

Phase 1 of the M.Tech. project has been completed successfully, delivering all three planned artefacts within the scheduled timeline:

1. A **literature review and benchmark survey** (separate deliverable) covering the ASVspoof challenge series (2015–2021), classical and deep learning anti-spoofing approaches, and robustness and domain-mismatch findings.
2. A curated **dataset of 17,947 utterances** from 10 distinct sources, pre-processed to 8 kHz/G.711 banking telephony conditions, with strict speaker-independent and system-independent splits, noise augmentation variants, and a fully held-out Microsoft Edge-TTS generalisation test set.
3. A modular, reproducible **feature extraction pipeline** producing 257-dimensional handcrafted acoustic feature vectors and 128-bin log-Mel spectrograms, with all hyperparameters centralised in `config.yaml` and dataset integrity validated by `diagnose.py`.

All six Phase 2 prerequisites are satisfied. Preliminary feature distribution analysis confirms that the chosen feature groups show clear inter-class separation consistent with the literature-predicted discriminative signals.

The project remains on schedule and on scope. No material risks or blockers are anticipated for Phase 2.

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